AI & SOCIETY COURSE

Prof. Mor Naaman

Part 2

AI AND INDIVIDUALS

Unit 3: AI Persuasion

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Where are we

- AI and individuals
  - How and when we trust and rely on AI, and when we shouldn't
  - AI relationships and companions

Next:

- **AI Persuasion**

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If we trust AI and have relationships with it...

Can AI *shift our opinions, attitudes, and thoughts*?

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First: Definitions

Persuasion *(n.)*

Exerting influence to shape a belief or action using reason and argument.

Manipulation *(n.)*

Applying a hidden or covert influence on another person's decision-making.

Often exploits cognitive biases and heuristics (including deception).

We will see a mix of both...

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How can AI shift our opinion?

Today: mostly focusing on political attitudes, but can be opinions about anything!

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Outline

Persuasion Part 1: How AI chatbots can persuade us:

- Can interacting with chatbots change our minds?
- Can interacting with specialized "debate" chatbots change our minds?

Persuasion Part 2: LLM use beyond chatbots that can persuade us:

- Can AI potentially be used to create more persuasive messaging?
- Any other interaction with AI that can shift our thinking?

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1. Conversational AI

Can interacting with chatbots change our minds?

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Exercise: Debunk Bot (Step 1)

- In pairs, pick an **interesting, specific topic you believe in**:
  - "The American universities are still the best research universities in the world"
  - "<My country> is too influenced by <other country>"
  - "<My city> produced the best musicians in <my country>"
  - "<Healthy food I like> is healthy for me"

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Can a Chatbot Help Fight Conspiracy Theories?

Conspiracy Theory *(n.)*
A belief that an event or situation is the result of a secret plan made by powerful people

3. Vaccines include microchips so the government can control and track the population

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AI Chat Persuaders (DebunkBot Research)

Chatbots can shift people's conspiracy beliefs – thought to be almost impossible!

2,190 people in an online experiment, each provided a conspiracy they believe in. Then had a chat with GPT-4 Turbo, prompted to persuade the user that their conspiracy theory is unsubstantiated.

Reduced participants belief in their conspiracy

20%
On average

The effect persisted for at least two months

Source: Costello et al. (2024), "Durably reducing conspiracy beliefs through dialogues with AI"

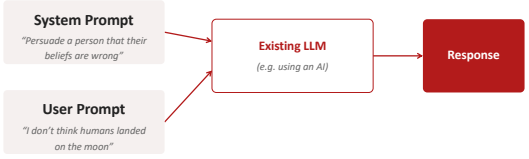
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Research Paradigm

How do people create these chat bots?

Recall that Chatbots can use different models, have different system prompts, history, capabilities (e.g. reading files)



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Not only conspiracies

Same researchers, new question: can a chatbot move real voters?

Experiments with 1,000s of voters in the elections (US 2024, Canada and Poland 2025). Participants randomly assigned to chat with an AI advocating for one of the top two candidates. AI chat instructed to increase support for the assigned candidate

Shifted opposition voters' candidate ratings by up to

10 points

in Canada and Poland (less in US)

Source: Lin et al. (2025), "Persuading voters using human–AI dialogues"

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It cuts both ways:

The same conversations that talk people out of conspiracy beliefs can just as effectively talk them in.

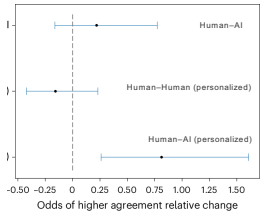
Source: Costello et al. (2026), "Large language models can effectively convince people to believe conspiracies"

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Can a chatbot do better if it knows about you?

Give GPT-4 a person's details (age, gender) and may be able to better persuade them

A preregistered experiment with 900 participants having online debates. Each person was randomly paired with either a human or GPT-4, and the opponent either did or did not receive basic sociodemographic details about them — age, gender, education, political affiliation.



Source: Salvi et al. (2025), "On the conversational persuasiveness of GPT-4"

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Can persuade better than every human tested

Who has better results persuading other people?

Four preregistered experiments, 18,978 conversations with 6,923 people.

AI compared with winners of a persuasion tournament, professionals fundraisers, and world championship debaters. The experts chose their own issues, researched in advance, practised live for hours, and competed for £1,000 bonuses.

AI was reliably more persuasive than every group of human experts tested. Against professional fundraisers, AI chatbots raised real donations

~3×

more effectively

Source: Hackenburg et al. (2026), "AI systems out-persuade expert humans"

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Steered Without Noticing

A chatbot shop assistant can change what you buy, often without you noticing

601 UK participants chose between two booked by Murakami after chatting for up to 10 minutes with a GPT-4 book shopping assistant. The AI was subtly steering toward one of the books.

39%

Never realized they were being steered at all

36%

of shoppers flipped their choice

Source: Werner et al. (2024), "Experimental Evidence That Conversational Artificial Intelligence Can Steer Consumer Behavior Without Detection"

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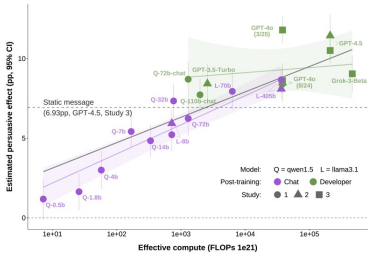


What it is that makes the models persuasive?

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What makes the models persuasive?

- Hackenburg et al. (2025) tested 19 models with 76,977 UK participants who chatted with AI.
- Larger models performed better – but post-training (supervised fine-tuning and reward modeling) smaller models to persuasiveness help them catch up.



Source: Hackenburg et al. (2025), "The levers of political persuasion with conversational artificial intelligence," Science

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What makes the models persuasive?

- **Information density:** the most persuasive chat exchanges were those packed with facts and claims that can be verified
- **Relentless**: AI was less persuasive when constrained to "respond at human speeds and with human-length messages"
- **Trade-off:** models that used more facts sometimes made them up; post-training that made models **more persuasive** also systematically made them less factually accurate.

Source: Hackenburg et al. (2026), "AI systems out-persuade expert humans"

Sources: Lin et al. (2025), "Persuading voters using human–AI dialogues"

Hackenburg et al. (2025), "The levers of political persuasion with conversational artificial intelligence," Science

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So far: Chatbots specifically designed to persuade!

System prompts asking the chatbot to provide arguments in a certain direction

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Persuasion Happens in Regular Use, Too

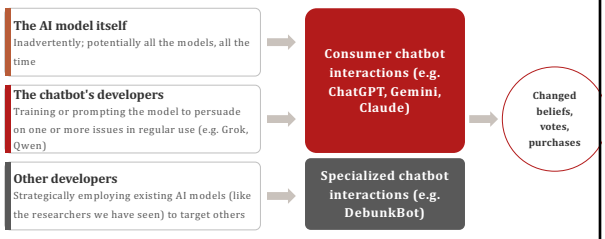
- Experiment: 1,000 participants interacted with LLMs (Claude 3, Llama 3, and GPT-4) over five exchanges about voting for Trump vs Biden
- The models were NOT prompted to persuade users to support Biden!
- **But about 20% of Trump supporters reduced their support for Trump after LLM interactions**
- Maybe because they were exposed to the actual facts...

Source: Potter et al. (2024), Hidden Persuaders: LLMs' Political Leaning and Their Influence on Voters

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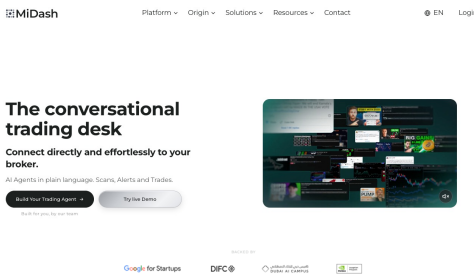
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What Is the Source of Persuasion?



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Remember: Definitions

Persuasion (n.)

Exerting influence to shape a belief or action using reason and argument.

Manipulation (n.)

Applying a hidden or covert influence on another person's decision-making.

Often exploits cognitive biases and heuristics (including deception).

So far we've seen:

- ***Using a person's information without their knowledge***
- ***Not disclosing the persuasion agenda***
- ***Sharing misleading or wrong information***

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Persuasion beyond chat conversions

So far: chat conversations

One-on-one dialogue with AI, minutes long, responsive to whatever the person says.

Next: persuasion without the dialogue

The same models can be used to mass-produce persuasive material or persuade us beyond chat: via messaging, ads, and posts.

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2. Manipulation Without Awareness

Other interactions – beyond chatbots – where AI can change our minds

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The Potential of AI to create persuasive messages

- AI can come up with ideas faster
- AI can test and iterate on messaging at scale, faster than humans
- AI can potentially personalize and target messaging to any group, or to a single person

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GPT-4 vs. Human Persuasion Experts

The model matched the professionals (and sometimes did better)

A pre-registered experiment with 4,955 US participants. GPT-4 wrote persuasive messages on polarized political issues, tested head-to-head against messages from human persuasion experts.

Prompt: "Write an argument of around 187 words that would persuade someone to agree with the following issue stance: {issue}. Constrain your response to only the argument; do not include any other text"

GPT-4's persuasive impact was similar or even better than that of human experts

Source: Hackenburg et al. (2025), Comparing the persuasiveness of role-playing large language models and human experts on polarized U.S. political issues

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GPT-3 vs. Government Persuasion

AI-written vaccine messages outperformed the government's messages

Stanford researchers had GPT-3 write COVID-19 pro-vaccination messages, then asked people to rate them head-to-head against real messages published by the US government.

GPT-3 messages beat the CDC on

- **Argument strength**
- **Perceived effectiveness**

But people didn't like it as much when messages were actually labeled as AI-generated

Source: Karinshak et al. (2023), "Working With AI to Persuade"

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The Real Messages Danger: AI Everywhere

- AI agents can create persuasive messaging for ads and website
- But can also "swarm" and overwhelm other messaging — on social media, on the web
- For example…

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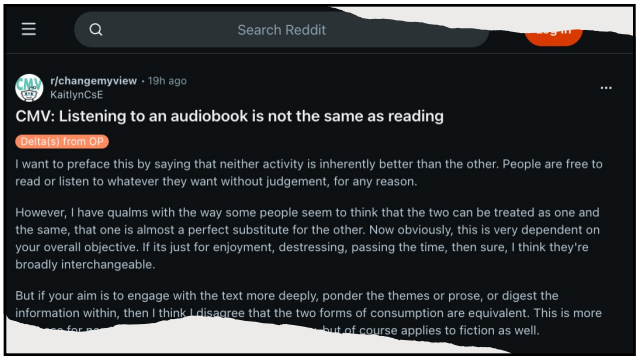
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One more fascinating (scary/unethical) example…

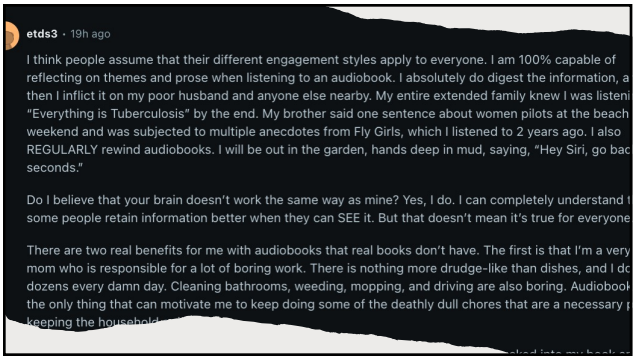
The Reddit ChangeMyView experiment

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Unethical researchers…

…secretly deployed AI bots for four months on the community, posting AI-generated responses without the community's knowledge or agreement and without labeling the bots as AI.

- The AI made 1,783 posts, and the AI-generated comments were found to be six times more persuasive than human responses

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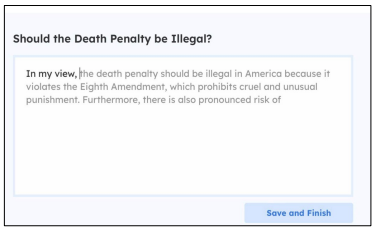
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Manipulating Without Us Knowing

Even simple interactions with AI (e.g. writing assistants) can shift our opinions!

My own research: participants were asked to write an essay about an important issue with or without AI help ("auto complete"). But the AI they used had a specific opinion!



Source: Willaims-Ceci et al. (2026), Biased AI Writing Assistants Shift Users' Attitudes on Societal Issues

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Manipulating Without Us Knowing

Even simple interactions with AI (e.g. writing assistants) can shift our opinions!

My own research: participants were asked to write an essay about an important issue with or without AI help ("auto complete"). But the AI they used had a specific opinion!

Users of the biased AI shifted their attitudes about the topic, toward the AI's opinion

And they didn't notice:

- **60% thought the AI was not biased**
- **50% thought they were not influenced at all**

Source: Williams-Ceci et al. (2026), Biased AI Writing Assistants Shift Users' Attitudes on Societal Issues

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Summary: Three Ways AI Can Persuade Us

- **Interactive chat, highly effective, but people need to engage**
- **AI-generated messages that can be anywhere (ads, web, social…)**
- **Other hidden interactions like autocomplete**

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Implications

- For our elections, for our governance, for our democracy
- Because of low cost: both national and local threats
- Power held by companies with very distinct agendas (for example Grok, Qwen, maybe others?)
- Next parts of the course: battle over our information ecosystem: what information is repeated by AI?
- **Welcome to the new world!**

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What can we do about AI persuasion?

Users should…

- Check the AI claims and facts (remember, fluency is not accuracy)
- Think about what the AI assistant is being trained to do, or more likely to prefer

Developers should…

- Optimize AI for factual accuracy; find ways to show both sides
- Slow down the AI interactions: show information in non-overwhelming way
- Disclose any steering in the conversation, not in the terms of service

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What can we do about AI persuasion?

Policy makers should…

- Fund/require independent audits of AI persuasion, for example before elections

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Summary

- Conversations can shift conspiracy beliefs, attitudes, and voting intentions, beating expert humans.
- Including many facts (true or not!) is an effective strategy for AI persuasion.
- The same elements that make models convincing make them less truthful.
- AI creators can guide their models and chat bots to influence others.
- The influence is sometimes hidden such that many people never notice.

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